

Safety Data Sheet

ULTRABOND TURF 2 STARS PRO /B

Safety Data Sheet dated: 21/06/2023 - version 2

Date of first edition: 28/05/2019



Section 1: Identification

GHS Product identifier

Mixture identification:

Trade name: ULTRABOND TURF 2 STARS PRO /B

Trade code: 906RC9999

Recommended use of the chemical and restrictions on use

Recommended use: Adhesive

Uses advised against: Data not available.

Supplier's details

Company: MAPEI AUSTRALIA Pty Ltd

180 Viking Drive Wacol QLD 4076 Australia

T. +61 7 32765000 (Mon-Fri 8am to 4.30pm)

F. +61 7 32765076

Responsible: sales@mapei.com.au

Emergency phone number

Australian Poisons Information Centre 24 Hour Service 13 11 26

Police or Fire Brigade 000

Section 2: Hazard(s) identification



Classification of the Hazardous chemical

Acute toxicity (inhalation), Category 4

Harmful if inhaled.

Skin irritation, Category 2

Causes skin irritation.

Eye irritation, Category 2A

Causes serious eye irritation.

Respiratory Sensitisation, Category 1

May cause allergy or asthma symptoms or breathing difficulties if inhaled.

Skin Sensitisation, Category 1

May cause an allergic skin reaction.

Carcinogenicity, Category 2

Suspected of causing cancer.

Specific target organ toxicity — single exposure, Category 3

May cause respiratory irritation.

Specific target organ toxicity — repeated exposure, Category 2

May cause damage to organs through prolonged or repeated exposure.

Adverse physicochemical, human health and environmental effects:

No other hazards

GHS label elements, including precautionary statements

Hazard pictograms and Signal Word



Danger

Hazard statements

H315 Causes skin irritation.

H317 May cause an allergic skin reaction.

H319 Causes serious eye irritation.

H332 Harmful if inhaled.

H334 May cause allergy or asthma symptoms or breathing difficulties if inhaled.

H335 May cause respiratory irritation.

H351 Suspected of causing cancer.

H373 May cause damage to organs through prolonged or repeated exposure.

Precautionary statements

P201 Obtain special instructions before use.

P202	Do not handle until all safety precautions have been read and understood.
P260	Do not breathe mist/vapours/spray.
P264	Wash hands thoroughly after handling.
P271	Use only outdoors or in a well-ventilated area.
P280	Wear protective gloves/clothing and eye/face protection.
P284	[In case of inadequate ventilation] wear respiratory protection.
P302+P352	IF ON SKIN: Wash with plenty of water.
P304+P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305+P351+P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P308+P313	IF exposed or concerned: Get medical advice/attention.
P312	Call a POISON CENTER if you feel unwell.
P314	Get medical advice/attention. if you feel unwell.
P321	Specific treatment (see supplementary instructions on this label)
P333+P313	If skin irritation or rash occurs: Get medical advice/attention.
P337+P313	If eye irritation persists: Get medical advice/attention.
P342+P311	If experiencing respiratory symptoms: Call a POISON CENTER or doctor/physician.
P362+P364	Take off contaminated clothing and wash it before reuse.
P403+P233	Store in a well-ventilated place. Keep container tightly closed.
P501	Dispose of contents/container in accordance with applicable regulations.

Other hazards which do not result in a classification

Other Hazards: No other hazards

Section 3: Composition and information on ingredients

Substances

no data available

Mixtures

Mixture identification: ULTRABOND TURF 2 STARS PRO /B

Hazardous components within the meaning of the "Australian Work Health and Safety (WHS)" regulation and related classification:

Qty	Name	Ident. Numb.	Classification	Registration Number
≥75 - <100 %	diphenylmethanediisocyanate isomers and homologues	CAS:9016-87-9 EC:618-498-9 Index:615-005-00-9	Acute Tox. 4, H332 Eye Irrit. 2A, H319 STOT SE 3, H335 Skin Irrit. 2, H315 Resp. Sens. 1, H334 Skin Sens. 1, H317 STOT RE 2, H373 Carc. 2, H351 Specific Concentration Limits: C ≥ 5%: Skin Irrit. 2 H315 C ≥ 5%: Eye Irrit. 2A H319 C ≥ 5%: STOT SE 3 H335 C ≥ 0.1%: Resp. Sens. 1 H334	
≥10 - <20 %	diphenylmethane-4,4'-diisocyanate	CAS:101-68-8 EC:202-966-0 Index:615-005-00-9	Acute Tox. 4, H332 Eye Irrit. 2A, H319 STOT SE 3, H335 Skin Irrit. 2, H315 Resp. Sens. 1, H334 Skin Sens. 1, H317 STOT RE 2, H373 Carc. 2, H351 Specific Concentration Limits: C ≥ 5%: Skin Irrit. 2 H315 C ≥ 5%: Eye Irrit. 2A H319 C ≥ 5%: STOT SE 3 H335 C ≥ 0.1%: Resp. Sens. 1 H334	01-2119457014-47-XXXX
≥10 - <20 %	o-(p-isocyanatobenzyl)phenyl isocyanate	CAS:5873-54-1 EC:227-534-9 Index:615-005-00-9	Carc. 2, H351 STOT RE 2, H373 Eye Irrit. 2A, H319 STOT SE 3, H335 Skin Irrit. 2, H315 Resp. Sens. 1, H334 Skin Sens. 1, H317 Acute Tox. 4, H332 Specific Concentration Limits:	01-2119480143-45-XXXX

C ≥ 5%: Skin Irrit. 2 H315
C ≥ 5%: Eye Irrit. 2A H319
C ≥ 5%: STOT SE 3 H335
C ≥ 0.1%: Resp. Sens. 1 H334

Section 4: First-aid measures

Description of necessary first-aid measures

In case of skin contact:

- Immediately take off all contaminated clothing.
- Remove contaminated clothing immediately and dispose of safely.
- After contact with skin, wash immediately with soap and plenty of water.

In case of eyes contact:

- After contact with the eyes, rinse with water with the eyelids open for a sufficient length of time, then consult an ophthalmologist immediately.
- Protect uninjured eye.

In case of Ingestion:

- Do not induce vomiting, get medical attention showing the SDS and the hazard label.

In case of Inhalation:

- If breathing is irregular or stopped, administer artificial respiration.
- In case of inhalation, consult a doctor immediately and show him packing or label.

Symptoms caused by exposure

Eye irritation
Eye damages
Skin Irritation
Erythema

Medical attention and special treatment

In case of accident or unwellness, seek medical advice immediately (show directions for use or safety data sheet if possible).

Section 5: Firefighting measures

Suitable extinguishing media

- None in particular.
- Water.
- Carbon dioxide (CO₂).

Extinguishing media which must not be used for safety reasons:

- None in particular.

Specific hazards arising from the chemical

- Do not inhale explosion and combustion gases.
- Burning produces heavy smoke.
- Hazardous combustion products: no data available
- Explosive properties: no data available
- Oxidizing properties: no data available

Special protective equipment and precautions for fire-fighters

- Use suitable breathing apparatus.
- Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
- Move undamaged containers from immediate hazard area if it can be done safely.

HazChem Code/Emergency Action code

N.A.

Section 6: Accidental release measures

Personal precautions, protective equipment and emergency procedures

- Wear personal protection equipment.
- Wear breathing apparatus if exposed to vapours/dusts/aerosols.
- Provide adequate ventilation.
- Use appropriate respiratory protection.

Environmental precautions

- Do not allow to enter into soil/subsoil. Do not allow to enter into surface water or drains.
- Limit leakages with earth or sand.

Methods and material for containment and cleaning up

- Suitable material for taking up: absorbing material, organic, sand
- Wash with plenty of water.

Section 7: Handling and storage

Precautions for safe handling

Avoid contact with skin and eyes, inhalation of vapours and mists.
Exercise the greatest care when handling or opening the container.
Do not use on extensive surface areas in premises where there are occupants.
Use localized ventilation system.
Don't use empty container before they have been cleaned.
Before making transfer operations, assure that there aren't any incompatible material residuals in the containers.
Contaminated clothing should be changed before entering eating areas.
Do not eat or drink while working.
See also section 8 for recommended protective equipment.

Conditions for safe storage, including any incompatibilities

Always keep in a well ventilated place.
Keep away from food, drink and feed.

Incompatible materials:

None in particular.

Instructions as regards storage premises:

Cool and adequately ventilated.

Section 8: Exposure controls and personal protection

Control parameters – exposure standards, biological monitoring

Community Occupational Exposure Limits (OEL)

	OEL Type	Country	Occupational Exposure Limit
diphenylmethanediisocyanate isomers and homologues CAS: 9016-87-9	ACGIH		Long Term: 0.05 ppm
	AUS	AUSTRALIA	Long Term: 0.02 mg/m ³ ; Short Term: 0.07 mg/m ³
	National	GERMANY	Long Term: 0.05 mg/m ³
diphenylmethane-4,4'-diisocyanate CAS: 101-68-8	National	SLOVENIA	Long Term: 0.05 mg/m ³ ; Short Term: 0.05 mg/m ³
	National	NORWAY	Long Term: 0.05 mg/m ³ - 0.005 ppm; Short Term: 0.01 ppm A 4
	National	SWEDEN	Ceiling - Long Term: 0.03 mg/m ³ - 0.002 ppm; Short Term: 0.05 mg/m ³ - 0.005 ppm SWEDEN, Ceiling limit value
	ACGIH		Long Term: 0.005 ppm Resp sens
	National	POLAND	Long Term: 0.03 mg/m ³ ; Short Term: 0.09 mg/m ³
	National	AUSTRIA	Long Term: 0.05 mg/m ³ - 0.005 ppm; Short Term: 0.1 mg/m ³ - 0.01 ppm
	ACGIH		Long Term: 0.005 ppm respiratory sensitization (listed under Methylene bisphenyl isocyanate (MDI))
	AUS	AUSTRALIA	Long Term: 0.02 mg/m ³ ; Short Term: 0.07 mg/m ³
	OSHA		Ceiling - Short Term: 0.2 mg/m ³ - 0.02 ppm
	National	SWEDEN	Long Term: 0.03 mg/m ³ - 0.002 ppm
	National	FRANCE	Long Term: 0.1 mg/m ³ - 0.01 ppm; Short Term: 0.2 mg/m ³ - 0.02 ppm
	National	SPAIN	Long Term: 0.052 mg/m ³ - 0.005 ppm
	National	DENMARK	Long Term: 0.05 mg/m ³ - 0.005 ppm
	National	GERMANY	Long Term: 0.05 mg/m ³
	National	PORTUGAL	Long Term: 0.005 ppm
	National	BELGIUM	Long Term: 0.052 mg/m ³ - 0.005 ppm
	National	CZECH REPUBLIC	Long Term: 0.05 mg/m ³
	National	HUNGARY	Long Term: 0.05 mg/m ³ ; Short Term: 0.05 mg/m ³
	National	ESTONIA	Long Term: 0.05 mg/m ³ - 0.005 ppm; Short Term: 0.1 mg/m ³ - 0.01 ppm
	National	CZECH	Ceiling - Short Term: 0.1 mg/m ³

REPUBLIC

National SLOVAKIA	Long Term: 0.002 mg/m ³
National SLOVAKIA	Long Term: 0.03 mg/m ³
National SLOVENIA	Long Term: 0.05 mg/m ³ ; Short Term: 0.05 mg/m ³
National ROMANIA	Short Term: 0.15 mg/m ³
National LITHUANIA	Long Term: 0.05 mg/m ³ - 0.005 ppm
National LITHUANIA	Ceiling - Short Term: 0.1 mg/m ³ - 0.01 ppm
ACGIH	Long Term: 0.005 ppm respiratory sensitization (listed under Methylene bisphenyl isocyanate (MDI))
OSHA	Ceiling - Short Term: 0.2 mg/m ³ - 0.02 ppm
National NORWAY	Long Term: 0.05 mg/m ³ - 0.005 ppm; Short Term: 0.01 ppm
National SLOVENIA	Long Term: 0.05 mg/m ³ - 0.005 ppm; Short Term: 0.05 mg/m ³ - 0.005 ppm

o-(p-isocyanatobenzyl)phenyl National GERMANY
isocyanate
CAS: 5873-54-1

National SLOVENIA Long Term: 0.05 mg/m³; Short Term: 0.05 mg/m³

Predicted No Effect Concentration (PNEC) values

diphenylmethane-4,4'-diisocyanate
CAS: 101-68-8

Exposure Route: Fresh Water; PNEC Limit: 1 mg/l

Exposure Route: Marine water; PNEC Limit: 0.1 mg/l

Exposure Route: Soil; PNEC Limit: 1 mg/kg

Exposure Route: Microorganisms in sewage treatments; PNEC Limit: 1 mg/l

Exposure Route: Intermittent release; PNEC Limit: 10 mg/l

o-(p-isocyanatobenzyl)phenyl
isocyanate
CAS: 5873-54-1

Exposure Route: Fresh Water; PNEC Limit: 1 mg/l

Exposure Route: Marine water; PNEC Limit: 0.1 mg/l

Exposure Route: Soil; PNEC Limit: 1 mg/kg

Exposure Route: Microorganisms in sewage treatments; PNEC Limit: 1 mg/l

Derived No Effect Level (DNEL) values

diphenylmethane-4,4'-diisocyanate
CAS: 101-68-8

Exposure Route: Human Dermal; Exposure Frequency: Short Term, systemic effects
Worker Industry: 50 mg/kg

Exposure Route: Human Inhalation; Exposure Frequency: Short Term, systemic effects
Worker Industry: 0.1 mg/m³

Exposure Route: Human Inhalation; Exposure Frequency: Short Term, local effects
Worker Industry: 0.1 mg/m³

Exposure Route: Human Inhalation; Exposure Frequency: Long Term, systemic effects
Worker Industry: 0.05 mg/m³

Exposure Route: Human Inhalation; Exposure Frequency: Long Term, local effects
Worker Industry: 0.05 mg/m³

Exposure Route: Human Dermal; Exposure Frequency: Short Term, systemic effects
Consumer: 25 mg/kg

Exposure Route: Human Inhalation; Exposure Frequency: Short Term, systemic effects
Consumer: 0.05 mg/m³

Exposure Route: Human Oral; Exposure Frequency: Short Term, systemic effects
Consumer: 20 mg/kg

Exposure Route: Human Inhalation; Exposure Frequency: Short Term, local effects
Consumer: 0.05 mg/m³

Exposure Route: Human Inhalation; Exposure Frequency: Long Term, systemic effects
Consumer: 0.025 mg/m³

Exposure Route: Human Inhalation; Exposure Frequency: Long Term, local effects
Consumer: 0.025 mg/m³

Exposure Route: Human Dermal; Exposure Frequency: Short Term, local effects
Worker Industry: 28.7 mg/cm²; Consumer: 17.2 mg/cm²

o-(p-
isocyanatobenzyl)phenyl
isocyanate
CAS: 5873-54-1

Exposure Route: Human Dermal; Exposure Frequency: Short Term, systemic effects
Worker Industry: 50 mg/kg; Consumer: 25 mg/kg

Exposure Route: Human Inhalation; Exposure Frequency: Short Term, systemic effects
Worker Industry: 0.1 mg/m³; Consumer: 0.05 mg/m³

Exposure Route: Human Dermal; Exposure Frequency: Short Term, local effects
Worker Industry: 28.7 mg/cm²; Consumer: 17.2 mg/cm²

Exposure Route: Human Inhalation; Exposure Frequency: Short Term, local effects
Worker Industry: 0.1 mg/m³; Consumer: 0.05 mg/m³

Exposure Route: Human Inhalation; Exposure Frequency: Long Term, systemic effects
Worker Industry: 0.05 mg/m³; Consumer: 0.025 mg/m³

Exposure Route: Human Inhalation; Exposure Frequency: Long Term, local effects
Worker Industry: 0.05 mg/m³; Consumer: 0.025 mg/m³

Exposure Route: Human Oral; Exposure Frequency: Short Term, systemic effects
Consumer: 20 mg/kg

Appropriate engineering controls

no data available

Individual protection measures, such as personal protective equipment (PPE)

Eye protection:

Use close fitting safety goggles, don't use eye lens.

Protection for skin:

Use clothing that provides comprehensive protection to the skin, e.g. cotton, rubber, PVC or viton.

Protection for hands:

Suitable materials for safety gloves; AS/NZS 2161.10:

Polychloroprene - CR: thickness $\geq 0,5\text{mm}$; breakthrough time $\geq 480\text{min}$.

Nitrile rubber - NBR: thickness $\geq 0,35\text{mm}$; breakthrough time $\geq 480\text{min}$.

Butyl rubber - IIR: thickness $\geq 0,5\text{mm}$; breakthrough time $\geq 480\text{min}$.

Fluorinated rubber - FKM: thickness $\geq 0,4\text{mm}$; breakthrough time $\geq 480\text{min}$.

Use protective gloves that provides comprehensive protection, e.g. P.V.C., neoprene or rubber.

Respiratory protection:

Respiratory protection must be used where exposure levels exceed workplace exposure limits. Refer to AS/NZS 1715-1716 for information on selection and use of appropriate respiratory protection equipment.

Use adequate protective respiratory equipment.

Section 9: Physical and chemical properties

Physical state: Liquid

Appearance: liquid

Color: light brown

Odour: Characteristic

pH: no data available

Melting point / freezing point: no data available

Initial boiling point and boiling range: no data available

Flash point: 229 °C (444 °F)

Evaporation rate: no data available

Flammability (Solid, Gas) no data available

Lower and upper explosion limit/flammability limits: no data available

Vapour pressure: no data available

Vapour density: no data available

Relative density: no data available

Solubility in water: no data available

Solubility in oil: no data available

Partition coefficient (n-octanol/water): no data available

Auto-ignition temperature: no data available

Decomposition temperature: no data available

Kinematic viscosity: no data available

Volatile Organic compounds - VOCs = No data available

Particle characteristics:

Particle size: no data available
Particle size distribution: no data available
Shape and aspect ratio: no data available
Specific surface area: no data available

Section 10: Stability and reactivity

Reactivity

Stable under normal conditions

Chemical stability

no data available

Possibility of hazardous reactions

None.

Conditions to avoid

Stable under normal conditions.

Incompatible materials

None in particular.

Hazardous decomposition products

None.

Section 11: Toxicological information

Information on toxicological effects

Toxicological Information of the Preparation

a) acute toxicity	The product is classified: Acute toxicity (inhalation), Category 4(H332)
b) skin corrosion/irritation	The product is classified: Skin irritation, Category 2(H315)
c) serious eye damage/irritation	The product is classified: Eye irritation, Category 2A(H319)
d) respiratory or skin sensitisation	The product is classified: Respiratory Sensitisation, Category 1(H334), Skin Sensitisation, Category 1(H317)
e) germ cell mutagenicity	Not classified Based on available data, the classification criteria are not met
f) carcinogenicity	The product is classified: Carcinogenicity, Category 2(H351)
g) reproductive toxicity	Not classified Based on available data, the classification criteria are not met
h) STOT-single exposure	The product is classified: Specific target organ toxicity — single exposure, Category 3(H335)
i) STOT-repeated exposure	The product is classified: Specific target organ toxicity — repeated exposure, Category 2(H373)
j) aspiration hazard	Not classified Based on available data, the classification criteria are not met

Toxicological information on main components of the mixture:

diphenylmethanediisocyanate isomers and homologues	a) acute toxicity	LD50 Oral Rat > 10000 mg/kg
		LD50 Skin Rabbit > 9400 mg/kg
		LC50 Inhalation Dust Rat = 0.31 mg/l 4h
		LD50 Skin Rabbit > 9.4 g/kg
		LC50 Inhalation Rat = 490 mg/m3 4h
		LD50 Oral Rat = 49 g/kg
	g) reproductive toxicity	NOAEL Inhalation Rat = 12 mg/m3
diphenylmethane-4,4'-diisocyanate	a) acute toxicity	LD50 Oral Rat > 2000 mg/kg
		LD50 Skin Rabbit > 9400 mg/kg
	b) skin corrosion/irritation	Skin Irritant Skin Rabbit Positive
	d) respiratory or skin sensitisation	Skin Sensitization Skin Mouse Positive
		Respiratory Sensitization Inhalation Positive

f) carcinogenicity	Carcinogenicity Inhalation Rat = 6 mg/m ³	2 y
g) reproductive toxicity	NOAEL Inhalation Rat = 12 mg/m ³	20 d
o-(p-isocyanatobenzyl)phenyl isocyanate		
a) acute toxicity	LD50 Skin Rabbit > 9400 mg/kg	
	LD50 Oral Rat > 2000 mg/kg	
e) germ cell mutagenicity	NOAEL Inhalation Rat = 12 mg/m ³	

Section 12: Ecological information

Ecotoxicity

Adopt good working practices, so that the product is not released into the environment.

Eco-Toxicological Information:

List of Eco-Toxicological properties of the product

Not classified for environmental hazards.

Based on available data, the classification criteria are not met

List of Eco-Toxicological properties of the components

Component	Ident. Numb.	Ecotox Data
diphenylmethanediisocyanate isomers and homologues	CAS: 9016-87-9 - EINECS: 618-498-9 - INDEX: 615-005-00-9	a) Aquatic acute toxicity : LC50 Fish > 1000 mg/L 96 a) Aquatic acute toxicity : EC50 Daphnia > 1000 mg/L 24 b) Aquatic chronic toxicity : NOEC Daphnia > 10 mg/L - 21 d a) Aquatic acute toxicity : EC50 Algae > 1640 mg/L 72 c) Bacteria toxicity : EC50 > 100 mg/L 3 d) Terrestrial toxicity : NOEC > 1000 mg/kg - 14 d e) Plant toxicity : NOEC > 1000 mg/kg - 14 d
diphenylmethane-4,4'-diisocyanate	CAS: 101-68-8 - EINECS: 202-966-0 - INDEX: 615-005-00-9	a) Aquatic acute toxicity : LC50 Fish > 1000 mg/L 96 a) Aquatic acute toxicity : EC50 Daphnia > 1000 mg/L 24 b) Aquatic chronic toxicity : NOEC Daphnia > 10 mg/L - 21 d a) Aquatic acute toxicity : EC50 Algae > 1640 mg/L 72 c) Bacteria toxicity : EC50 > 100 mg/L 3 d) Terrestrial toxicity : NOEC > 1000 mg/kg - 14 d e) Plant toxicity : NOEC > 1000 mg/kg - 14 d
o-(p-isocyanatobenzyl)phenyl isocyanate	CAS: 5873-54-1 - EINECS: 227-534-9 - INDEX: 615-005-00-9	a) Aquatic acute toxicity : LC50 Fish > 1000 mg/L 96 a) Aquatic acute toxicity : EC50 Daphnia > 1000 mg/L 24 b) Aquatic chronic toxicity : NOEC Daphnia > 10 mg/L - 21 d a) Aquatic acute toxicity : EC50 Algae > 1640 mg/L 72 c) Bacteria toxicity : EC50 > 100 mg/L 3 d) Terrestrial toxicity : NOEC > 1000 mg/kg - 14 d e) Plant toxicity : NOEC > 1000 mg/kg - 14 d

Persistence and degradability

no data available

Bioaccumulative potential

no data available

Mobility in soil

no data available

Other adverse effects
no data available

Section 13: Disposal considerations

Disposal methods

The generation of waste should be avoided or minimized wherever possible. Recover if possible.
no data available
Disposal of this product, solutions, packaging and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements.
Dispose of surplus and nonrecyclable products via a licensed waste disposal contractor.
Do not dispose of waste into sewers.

Disposal considerations:

Do not allow to enter drains or watercourses.
Dispose of product according to all federal, state and local applicable regulations.
If this product is mixed with other wastes, the original waste product code may no longer apply and the appropriate code should be assigned.
Dispose of containers contaminated by the product in accordance with local or national legal provisions. For further information, contact your local waste authority.

Special precautions:

This material and its container must be disposed of in a safe way. Care should be taken when handling untreated empty containers.
Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.
Empty containers or liners may retain some product residues. Do not re-use empty containers.

Section 14: Transport information

Not classified as dangerous in the meaning of transport regulations.

UN number

no data available

UN proper shipping name

no data available

Transport hazard class(es)

no data available

Packing group, if applicable

no data available

Environmental hazards

no data available

Special precautions for user

ADG-Subsidiary hazards no data available
ADG-S.P.: no data available

Road and Rail (ADR-RID):

no data available
ADR-Hazard identification number: NA

Air (IATA):

no data available

Sea (IMDG):

no data available

Additional Information

no data available

HazChem Code/Emergency Action code

no data available

Section 15: Regulatory information

Safety, health and environmental regulations specific for the product in question

This Safety Data Sheet has been prepared according to the Australian Work Health and Safety (WHS) act and the Code of Practice on preparation of safety data sheets for Hazardous Chemicals.
AICIS: all components are listed

Section 16: Any other relevant information

Code	Description
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.

H319	Causes serious eye irritation.
H332	Harmful if inhaled.
H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H335	May cause respiratory irritation.
H351	Suspected of causing cancer.
H373	May cause damage to organs through prolonged or repeated exposure.
H373	May cause damage to organs through prolonged or repeated exposure if inhaled.

Code	Hazard class and hazard category	Description
3.1/4/Inhal	Acute Tox. 4	Acute toxicity (inhalation), Category 4
3.2/2	Skin Irrit. 2	Skin irritation, Category 2
3.3/2A	Eye Irrit. 2A	Eye irritation, Category 2A
3.4.1/1	Resp. Sens. 1	Respiratory Sensitisation, Category 1
3.4.2/1	Skin Sens. 1	Skin Sensitisation, Category 1
3.6/2	Carc. 2	Carcinogenicity, Category 2
3.8/3	STOT SE 3	Specific target organ toxicity — single exposure, Category 3
3.9/2	STOT RE 2	Specific target organ toxicity — repeated exposure, Category 2

This document was prepared by a competent person who has received appropriate training.

Main bibliographic sources:

ECDIN - Environmental Chemicals Data and Information Network - Joint Research Centre, Commission of the European Communities

SAX's DANGEROUS PROPERTIES OF INDUSTRIAL MATERIALS - Eight Edition - Van Nostrand Reinold

The information contained herein is based on our state of knowledge at the above-specified date. It refers solely to the product indicated and constitutes no guarantee of particular quality.

It is the duty of the user to ensure that this information is appropriate and complete with respect to the specific use intended.

This SDS cancels and replaces any preceding release.

Legend to abbreviations and acronyms used in the safety data sheet:

ACGIH: American Conference of Governmental Industrial Hygienists

ADR: European Agreement concerning the International Carriage of Dangerous Goods by Road.

AND: European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways

ATE: Acute Toxicity Estimate

ATEmix: Acute toxicity Estimate (Mixtures)

BCF: Biological Concentration Factor

BEI: Biological Exposure Index

BOD: Biochemical Oxygen Demand

CAS: Chemical Abstracts Service (division of the American Chemical Society).

CAV: Poison Center

CE: European Community

CLP: Classification, Labeling, Packaging.

CMR: Carcinogenic, Mutagenic and Reprotoxic

COD: Chemical Oxygen Demand

COV: Volatile Organic Compound

CSA: Chemical Safety Assessment

CSR: Chemical Safety Report

DMEL: Derived Minimal Effect Level

DNEL: Derived No Effect Level.

DPD: Dangerous Preparations Directive

DSD: Dangerous Substances Directive

EC50: Half Maximal Effective Concentration

ECHA: European Chemicals Agency

EINECS: European Inventory of Existing Commercial Chemical Substances.

ES: Exposure Scenario

GefStoffVO: Ordinance on Hazardous Substances, Germany.

GHS: Globally Harmonized System of Classification and Labeling of Chemicals.

IARC: International Agency for Research on Cancer

IATA: International Air Transport Association.

IATA-DGR: Dangerous Goods Regulation by the "International Air Transport Association" (IATA).

IC50: half maximal inhibitory concentration

ICAO: International Civil Aviation Organization.

ICAO-TI: Technical Instructions by the "International Civil Aviation Organization" (ICAO).

IMDG: International Maritime Code for Dangerous Goods.

INCI: International Nomenclature of Cosmetic Ingredients.
IRCCS: Scientific Institute for Research, Hospitalization and Health Care
KAFH: KAFH
KSt: Explosion coefficient.
LC50: Lethal concentration, for 50 percent of test population.
LD50: Lethal dose, for 50 percent of test population.
LDLo: Leathal Dose Low
N.A.: Not Applicable
N/A: Not Applicable
N/D: Not defined/ Not available
NA: Not available
NIOSH: National Institute for Occupational Safety and Health
NOAEL: No Observed Adverse Effect Level
OSHA: Occupational Safety and Health Administration
PBT: Persistent, Bioaccumulative and Toxic
PGK: Packaging Instruction
PNEC: Predicted No Effect Concentration.
PSG: Passengers
RID: Regulation Concerning the International Transport of Dangerous Goods by Rail.
STEL: Short Term Exposure limit.
STOT: Specific Target Organ Toxicity.
TLV: Threshold Limiting Value.
TWATLV: Threshold Limit Value for the Time Weighted Average 8 hour day. (ACGIH Standard).
vPvB: Very Persistent, Very Bioaccumulative.
WGK: German Water Hazard Class.

*** Sheet model entirely changed in compliance to regulatory update.**